

Nathaniel DeLeon

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Portfolio: <https://completecanoe98.github.io/#>

EDUCATION

South Texas College

Graduation: December 2022

Associate of Science in Interdisciplinary Studies

GPA: 4.0 / 4.0

The University of Texas Rio Grande Valley

Expected Graduation: December 2026

Bachelor of Science in Computer Science

GPA: 4.0 / 4.0

SKILLS

- **Languages:** Python, C++, Dart
- **Libraries/Frameworks:** PyTorch, OpenCV, Pandas, RetinaFace, Ros, Gazebo, PyQt6, YoloWorld, Flutter
- **Tools:** Git/GitHub, Linux, VS Code, SolidWorks, Latex

WORK EXPERIENCE

Computer Science Department Lab Consultant – Edinburg Texas.

September 2024 – May 2025

- Coordinated daily lab operations to support Computer Science department faculty and students.
- Enforced necessary procedures to allow for proper success of students within the department who need the lab's assistance.
- Troubleshooting of labs equipment, when necessary, in cases where equipment finds itself to be faulty.

Research Assistant for Department of Computer Science – Edinburg Texas

May 2025 – August 2025

- Focused within Algorithm and Self Assembly, specifically studying topics such as Self Assembly and Drainability
- Collaborated with faculty to analyze algorithmic self-assembly complexity and probabilistic bounds.

Teaching Assistant for Department of Computer Science – Edinburg Texas

August 2025 – Current

- Acting Teaching Assistant for Automata and formal Languages Course. Previously taught Software Engineering 1 and Computer Science 1
- Assisted Professor in class procedures such as management of lab sessions.
- Provide Availability to students for individual or group guidance within subject topics such as desktop-based software and web-based software and their applications.
- Grade and provide feedback within Quiz assignments

ORGANIZATIONS

Telemetry Lead - Society of Aerospace and Robotic Engineers – Edinburg, Texas

August 2025 - Current

- Secured a \$15,000 prize as a finalist in the NASA ORBIT Competition by managing electronic and software project components.
- Managed the electronic and software components needed for project completion
- Developed project components validating the use of Visco Acoustic Electrospinning for Nanofiber Guidance.
- Delegated Team tasks to Telemetry members. These tasks contributed to the completion and success of the project.

PROJECTS

UAV Natural Language Tracker – Edinburg Texas

December 2025 – Current

- Built a python-based UAV targeting application within gazebo simulation that utilizes natural language vocabulary and reliable video streaming (UDP Based transmission).
- Utilized GPT 5.2 and YOLO World to develop a natural language detection pipeline, utilizing GPT's semantic reasoning and Yolo World's object detection
- Implemented full autonomous target tracking by mapping natural language text commands directly to real-time bounding boxes.

Remote Desktop System – Edinburg Texas

October 2025 –February 2026

- Built a python based low-latency remote desktop application that utilizes auto-discovery and reliable video streaming (UDP Based transmission).
- Reduced latency and bandwidth using threaded **OpenCV** processing and dynamic frame-dropping.
- Implemented full remote input control (mouse/keyboard) using **pynput** and **pyautogui**.

Deepfake Detector – Edinburg Texas

September 2025 – December 2025

- Engineered a hybrid spatial/temporal deep learning model combining **ResNet50** and **LSTMs** to detect video manipulation, utilizing transfer learning for robust feature extraction.
- Developed an automated video preprocessing pipeline using **RetinaFace** to detect, crop, and normalize face sequences from raw footage for consistent model input.
- Implemented a custom **PyTorch** training loop with dynamic frame sampling and validation checkpoints to optimize classification accuracy between real and fake media.

AWARDS

- UTRGV Luminary Scholar
- 6x President's List Award Recipient
- NASA ORBIT Competition Finalists